Romanization or resolution? A shuffled-label control for Cherokee–English translation memory

Transliteration can improve low-resource language technology by exposing cross-script or phonological overlap (Goyal et al. 2020; Moosa et al. 2023). But a raw-script–romanized comparison changes two things at once. It changes the linguistic labels, and it expands one source symbol into several Latin characters. The latter alone alters the geometry of character n-gram models. We isolate these factors in Cherokee–English translation, where the Cherokee syllabary provides an unusually clean test: a source symbol has a conventional Latin syllable label, but it can also be assigned an arbitrary label of the same length. A public corpus experiment finds a significant +1.02 chrF2 gain from romanization over raw syllabary. Crucially, length-matched shuffled labels reproduce about 80% of that gain. Thus the usual two-condition comparison overstates the evidence that transliteration helps because it encodes linguistically appropriate sub-symbol structure.

Background. Cherokee is polysynthetic and severely endangered. ChrEn provides 14K Cherokee–English pairs and separate in- and out-of-domain evaluations (Zhang et al. 2020). Work with Cherokee speakers also emphasizes that language technology must be developed through community partnership rather than treating an endangered language as a benchmark alone (Zhang et al. 2022). Our target is therefore methodological: we audit a common modeling inference, not propose a new community orthography. The romanization below is an automatically reproducible diagnostic based on Unicode character labels.

Method. We use the published ChrEn splits: 11,639 training, 1,000 development, 1,000 in-domain test, and 256 out-of-domain test pairs. To make the representational effect transparent, the translation system is one-nearest-neighbor translation memory. A TF–IDF vector represents each Cherokee sentence with character n-grams; the English side of its nearest training neighbor is returned as the translation. Native-script and romanized n-gram ranges are independently selected on development data from 1–4, 1–6, 2–5, 2–6, and 2–8. Both select 1–4. Output is scored with corpus chrF2 over character 1–6-grams (Popovic 2015).

The RAW condition uses Cherokee syllabary characters. The ROMANIZED condition maps each of 85 Cherokee symbols losslessly to the lowercase label in its Unicode character name (for example, a syllabary symbol labeled *NA* becomes `na`). This exposes within-syllable Latin substrings to the retriever. The key control randomly permutes these 85 labels among symbols *within label-length groups*. Each control therefore preserves the romanized alphabet, label frequencies, and every symbol's expansion length, while destroying the association between a symbol and its intended syllable label. We run 20 permutations. Uncertainty for the raw versus romanized difference comes from 2,000 paired bootstrap samples of test items.

Results. Table 1 shows that romanization raises in-domain chrF2 from 34.74 to 35.76, a +1.02 gain whose bootstrap 95% interval is $[0.49, 1.51]$; none of 2,000 resamples is non-positive. Source-side nearest- neighbor cosine similarity also rises from .344 to .443. The apparent benefit therefore replicates what an ordinary raw–romanized comparison would call a successful transliteration intervention.

Table 1: Cherokee-to-English translation-memory retrieval.

Representation	In-domain chrF2	Out-domain chrF2	Source cosine
Raw syllabary	34.74	18.54	.344
Unicode-label romanization	35.76	19.21	.443
Shuffled labels (20; mean ± SD)	$35.55 \pm .15$	–	–

The shuffled-label result changes the interpretation. Its scores range from 35.39 to 35.92; two of 20

arbitrary mappings outperform the linguistically labeled mapping. The romanized condition is only 0.21 above the shuffle mean, while the shuffle mean is 0.81 above raw. On this operationalization, roughly 80% of the headline gain is recoverable without correct syllable labels.

Analysis. A Cherokee syllabary character is atomic to the raw character model. Expanding it into multiple Latin characters creates additional partial matches, longer contexts, and more gradations of similarity. A random length-matched label does the same. Consequently, a raw–romanized contrast conflates a linguistic claim (the correct sub-symbol correspondence matters) with a resolution claim (more character positions help). The shuffle test supplies the missing counterfactual. The 0.21 advantage over the control mean, and the fact that 90% of controls score lower, leave room for a small linguistically structured effect, but 20 permutations do not distinguish it decisively.

Romanization also does not remove the corpus's domain problem. Both systems lose about 16 chrF2 points out of domain; the romanized advantage there is only 0.67. This echoes the original ChrEn finding that out-of-domain Cherokee translation is substantially harder. More broadly, transliteration is not a theory-neutral preprocessing step. It chooses which internal similarities a model may see and can privilege segmental Latin structure over syllabary-based orthographic practice. Claims of phonological or cross-script benefit should therefore include matched nonlinguistic recodings, report domain transfer, and ultimately be validated with speaker-approved representations and human judgment. The proposed control is cheap, model-agnostic, and applicable to any low-resource setting in which transliteration expands source symbols.

References

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